

Maag

Modular Agentic Autonomous Greenhouse

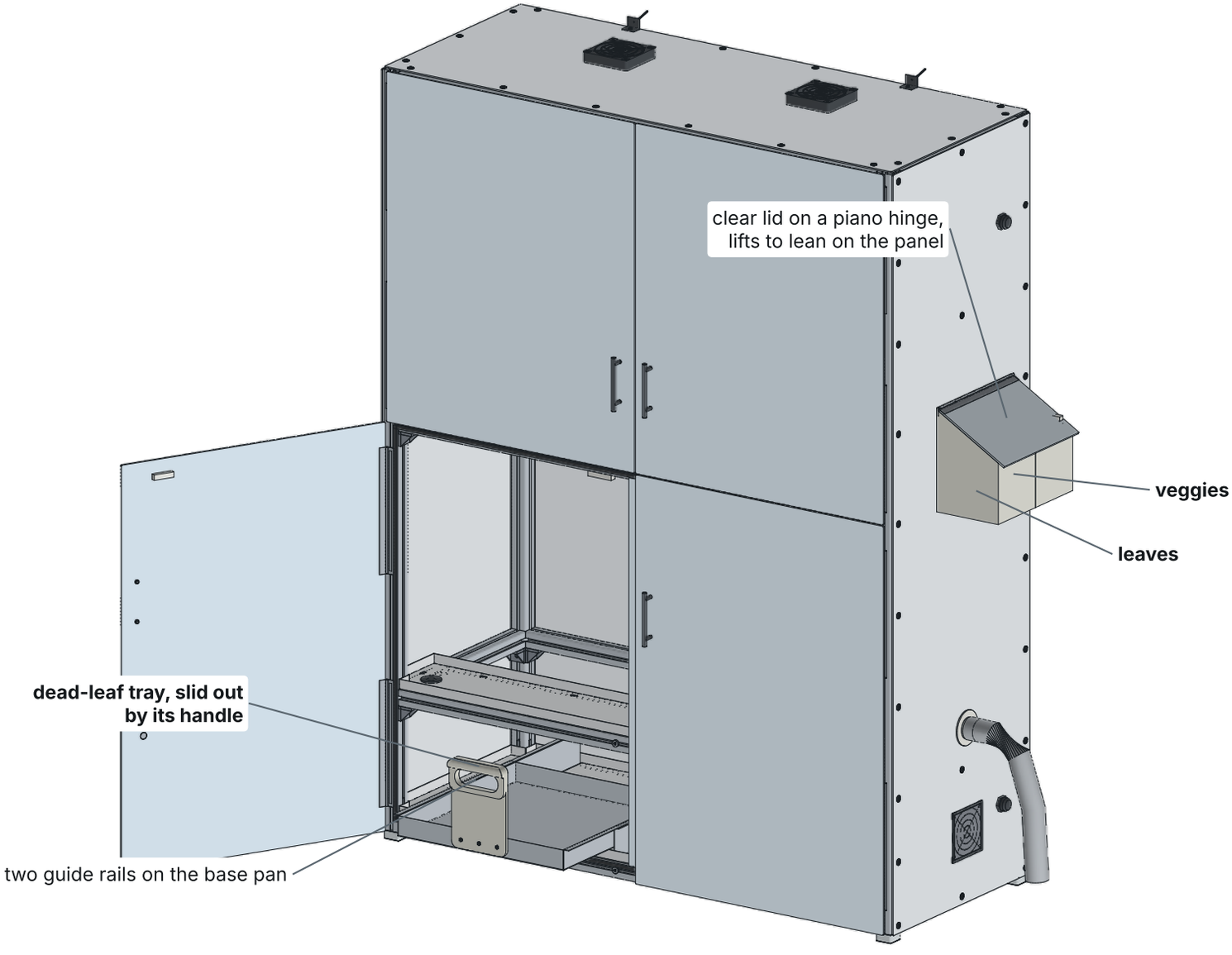
11 steps · tools · where every cut goes

Harvest assembly instructions, Harvest Rev A. The dead-leaf tray in the base, the side feeder on the right.

Design 0.6.0. The Maag cover render, the same on every booklet. Every drawing inside comes from harvest.py, enclosure.py and cabinet.py; the cut files come from the same model.

Harvest · where every cut goes

Dead leaves down into a tray in the base. Leaves and veggies out, into a feeder on the right.



What goes where

Leaves

Herb leaves to eat: out through the right panel, the front compartment.

Veggies

Fruit and pods: the back compartment. Each has its own lift-out tray.

Dead leaves

Down, into the 10 L tray in the base. Five drop points spread them along it.

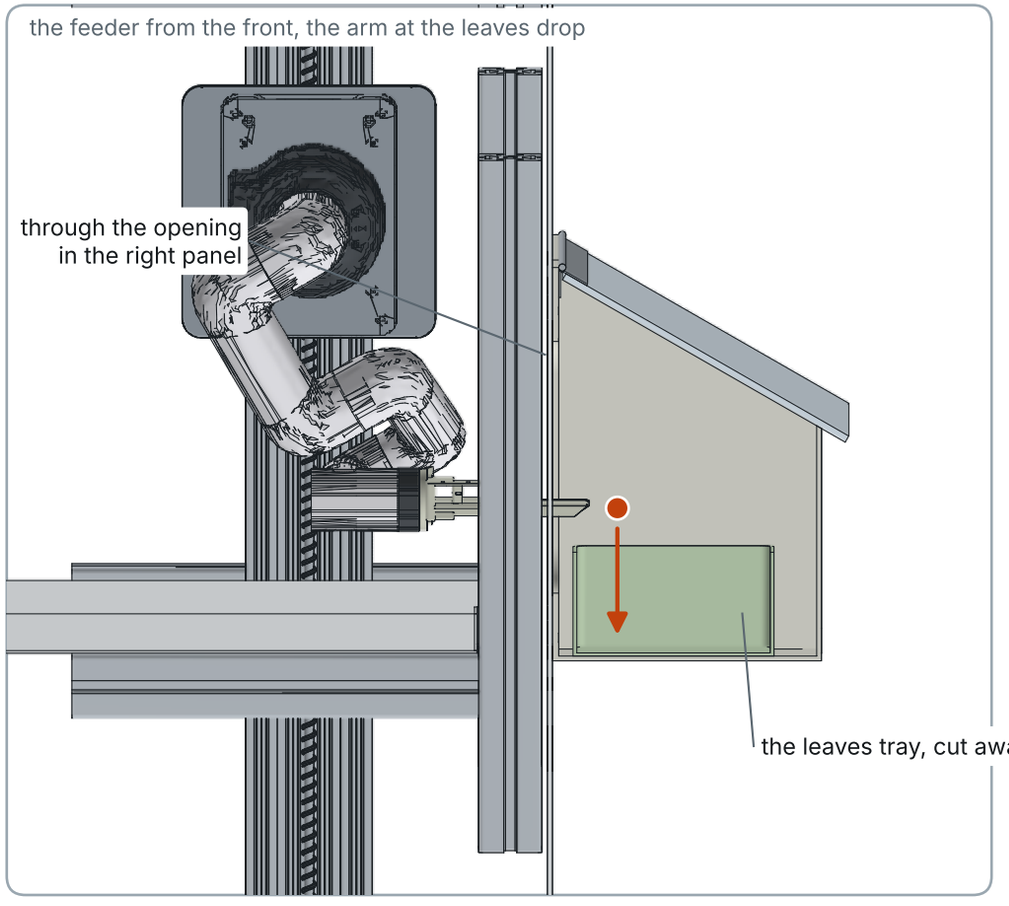
Collect

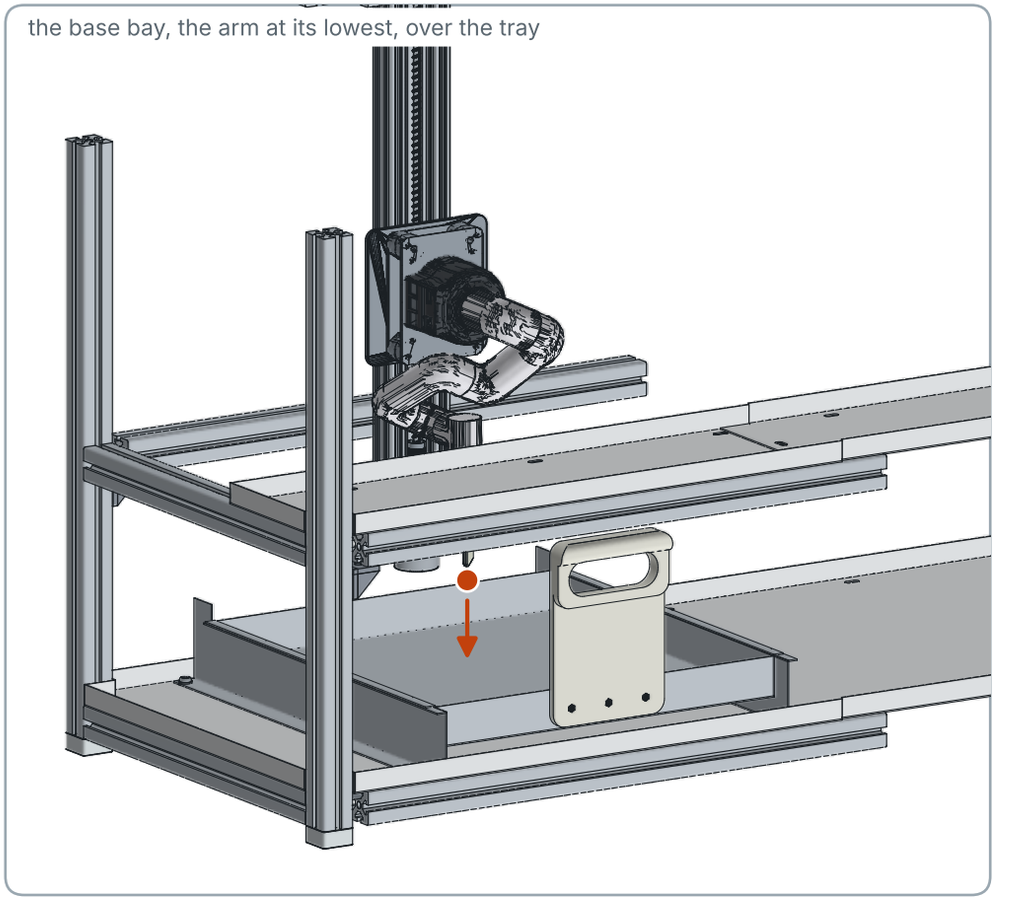
Lift the lid, lift a tray out. The arm has no power while the lid is open.

The feeder is two printed boxes side by side on the outside of the right panel at tier B, the third shelf up from the floor, with one clear lid over both, like the side feeders on a bird cage: the arm puts the harvest in from the inside, you take it out from the outside, and the doors stay shut. The tray in the base slides out of the bottom-left door alone. The modular cabinet can put the feeder on the left: it is the same parts, mirrored.

How the arm drops

The arm, the snip and every margin from reach.py: posed, then checked against the model.





Tightest gaps, mm (reach.py, the snip's printed body and the arm's own meshes)

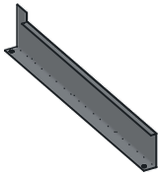
leaves	inside	plants (y < 210) 13	panel 29
leaves	through	plants (y < 210) 13	servo inside 17
leaves	release	plants (y < 210) 13	servo inside 17
veggies	inside	gantry 7	plants (y < 210) 13
veggies	through	plants (y < 210) 13	servo inside 17
veggies	release	plants (y < 210) 13	feeder floor 17

tray drop1	x	232	gantry 11	plants A (y < 210) 13
tray drop2	x	316	gantry 11	plants A (y < 210) 13
tray drop3	x	400	gantry 11	plants A (y < 210) 13
tray drop4	x	484	gantry 11	plants A (y < 210) 13
tray drop5	x	568	gantry 11	plants A (y < 210) 13

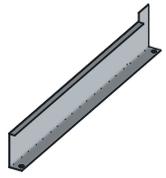
Each drop is three moves at the feeder: the snip level inside the cabinet, out through its compartment's opening above the tray, then down to 28 over the tray's rim and open. The snip's wide servo end never leaves the cabinet. Dead leaves go at the arm's lowest, jaws down, 55 over the rim. The poses are in drops.yaml for the Herb Snip's software; commissioning jogs to each before the first harvest.

Harvest · parts

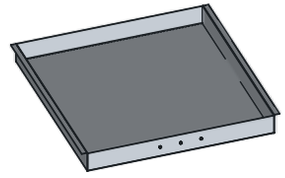
Count everything first. SendCutSend cuts and bends the metal and the lid; you print the rest.



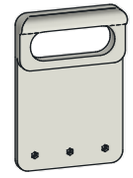
Guide rail L, 5052
1×



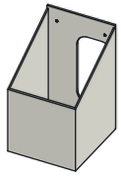
Guide rail R, 5052
1×



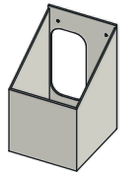
Dead-leaf tray, 5052
1×



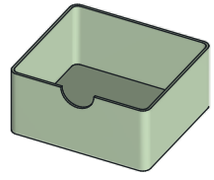
Handle, printed
1×



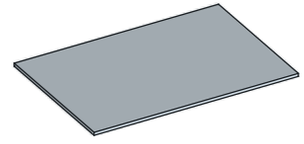
Feeder module, leaves
1×



Feeder module, veggies
1×



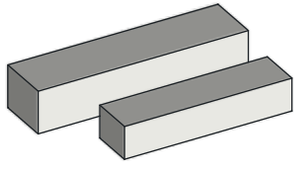
Feeder tray, printed
2×



Lid, clear 4.5
1×



Piano hinge, 1 ft
1×



Reed switch 7939WG
1×



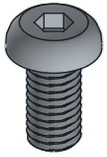
Rubber well nut M4
4×

Also, no picture needed:

VHB 4905, 1/2 in: both hinge leaves (from the enclosure's roll)
Two lever connectors (Wago 221-412) and 3 m of 2-core alarm cable for the switch
Six small adhesive cable clips for the switch's wire inside the cabinet
Print settings: PETG or PLA, 0.2 layers, 4 walls, 20 % gyroid; modules and trays standing on their floors, the handle flat on its back

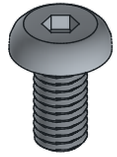
Harvest · fasteners

Every screw in the build. Lay them out in labelled pots first.



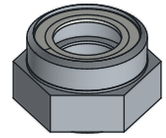
M8 × 16 button head · rails, the pan's own washers

4×



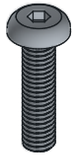
M5 × 10 button head · handle

3×



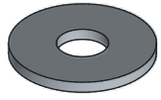
M5 nyloc · handle, in its pockets

3×



M5 × 20 button head · feeder, rail

4×



DIN 9021 washer M5 · the panel's

4×



M4 × 20 button head · feeder, top

4×



M4 rubber well nut, 10 hole

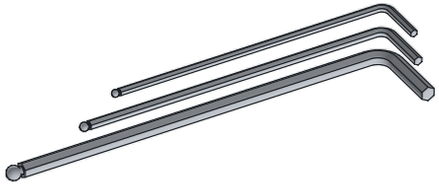
4×

Rails: the pan's bolts at x 150 and 650, front and back, go from M8 × 12 to M8 × 16: the rail's flange adds 1.6. Stack 4.8, reach 11.2, 1.05 short of the slot floor, as on the pan's UP piece. Feeder: M5 × 20 through the back plate, the panel and the tape: stack 11.875, 3.83 past the lips, 4.12 short of the floor, the same as every panel screw. The four M5 × 16 the panel wore there until now go back in the spares pot.

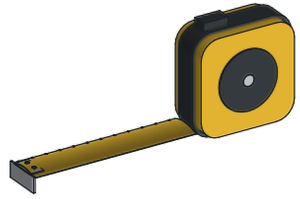
Well nuts are rubber sleeves with a brass nut moulded in the far end: push one into a 10 mm hole, turn the screw, and the sleeve folds up behind the panel and grips it. Nothing to hold on the inside, which is why the feeder's top screws use them: there is no frame behind the panel there.

Tools

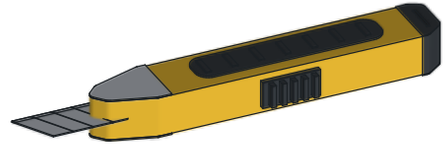
Everything the 11 steps need, and your printer for the five printed parts.



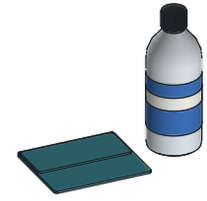
Hex keys, ball end
2.5 · 3 · 5 mm, long series
steps: 5: the rails' M8s, 2 · 3: every M5, 3, 7, 8 · 2.5: the M4s, 7, 8



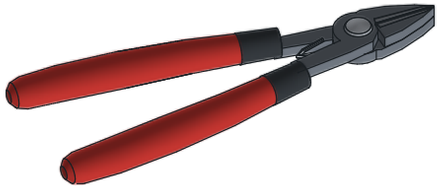
Tape measure
5 m
steps: the magnet's place on the lid, 5



Utility knife
18 mm snap-off, and scissors
steps: VHB, 5 and 9



Isopropyl alcohol
and microfibre cloths
steps: every VHB bond, 5 and 9



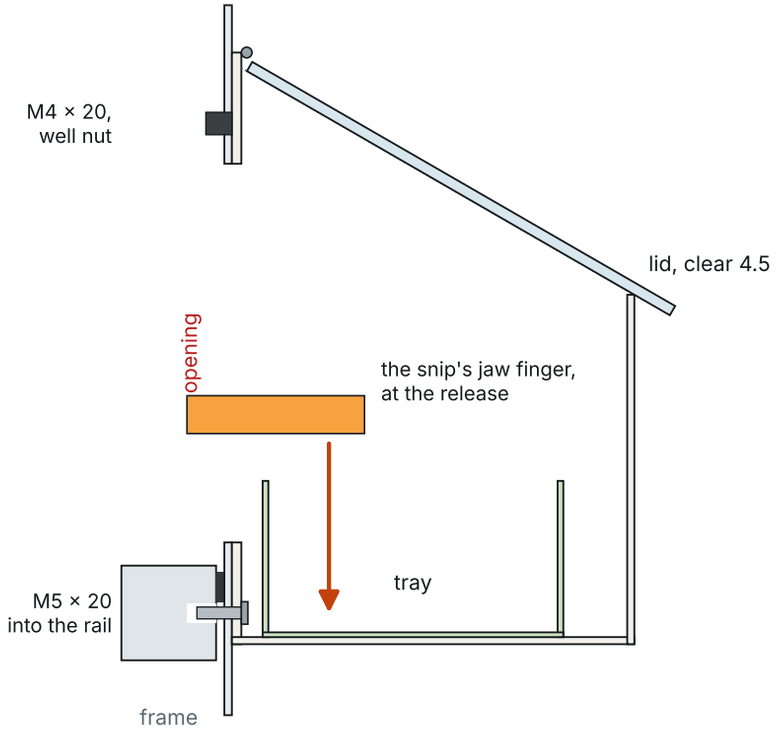
Side cutters
125 mm, flush cut
steps: the switch's cable, 10

And a wire stripper for the switch's cable, a step stool for the feeder, and a helper to hold each module while you start its screws.

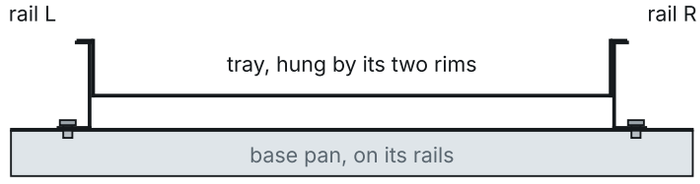
In section, to scale

Through the leaves compartment, across the dead-leaf tray, and through its handle.

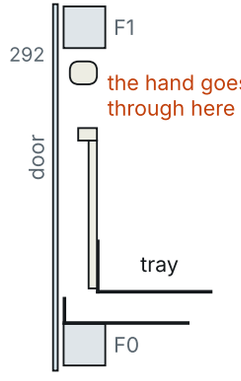
the feeder at the leaves drop, from the front



across the tray, from the front



through the handle, from the left



The tray's floor rides 5.0 over the pan's front lip, so it slides straight out; its rims and the rails' stops are 26.0 under the gantry's lowest part. The handle stops 12.8 under F1 and 11.63 behind the door's inside face. At the feeder the opening's sill is 9.8 over tier B's rail and the tray's rim 26.0 above the sill: nothing dropped can fall back into the cabinet.

1 Before you start

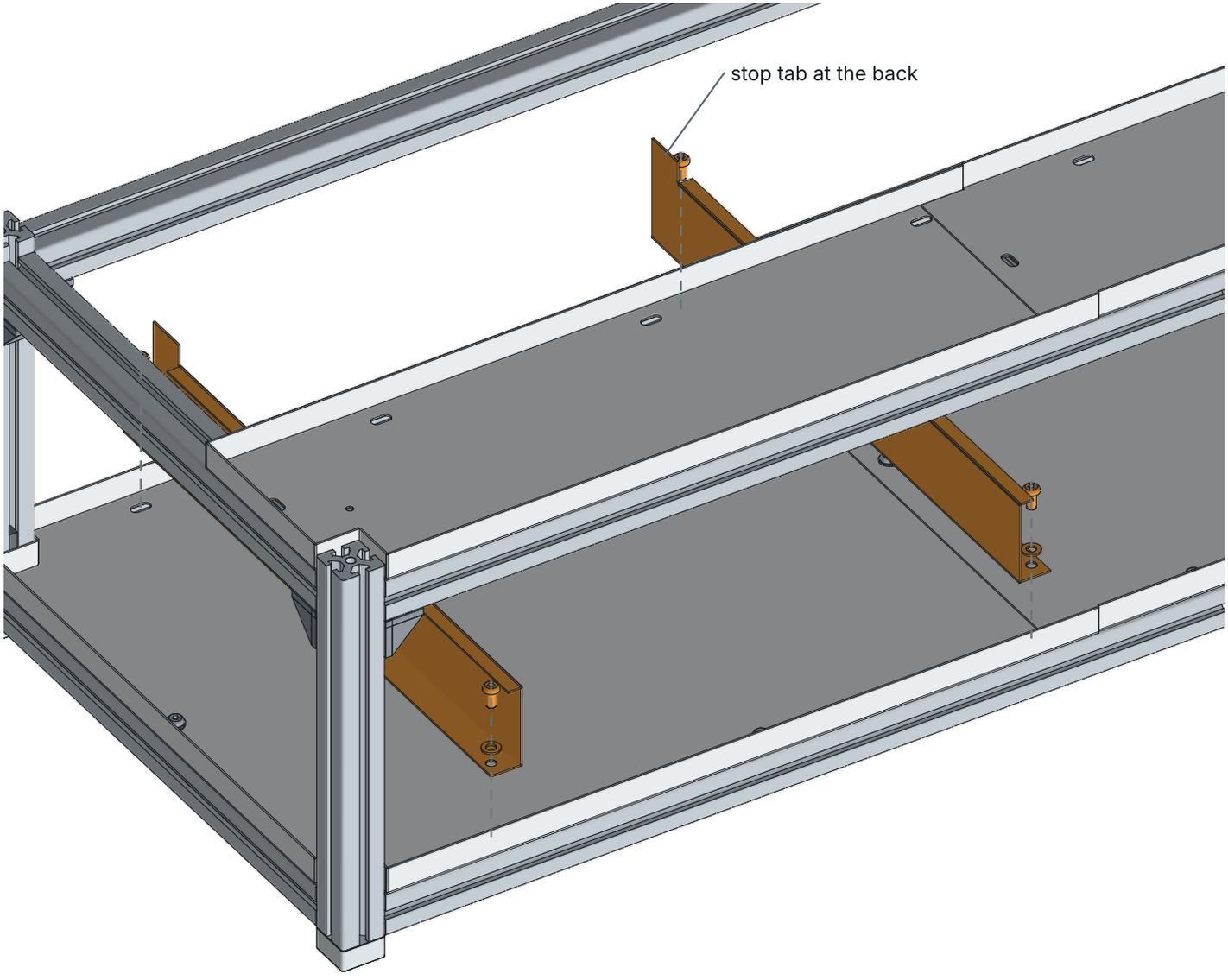
The cabinet finished to the enclosure booklet, the parts printed and cut. Six checks.

- [ ] **Enclosure Rev A done**
  Step 18 signed off, with the right panel that has the feeder's two openings and its four green screws on tier B's rail.
- [ ] **Robot off**
  Gantry homed and parked left, the E-stop pressed. Nothing in this booklet needs it powered.
- [ ] **Metal in**
  From SendCutSend: guide rails L and R, the dead-leaf tray (5052, bent), the clear lid. Check the rails mirror each other.
- [ ] **Printed**
  Handle, two feeder modules (leaves, veggies), two feeder trays: PETG or PLA, 4 walls, 20 %; no supports needed.
- [ ] **Fasteners**
  Against the fasteners page: four M8 × 16, three M5 × 10 and nylocs, four M5 × 20, four M4 × 20 and well nuts.
- [ ] **The switch**
  One more 7939WG reed switch and its magnet (as on the doors), two lever connectors, 3 m of 2-core cable.

Work in this order: the rails and the tray in the base first, then the feeder on the right panel. The tray needs only the bottom-left door open; the feeder is all done from outside, except the switch's cable, which runs inside the cabinet to the utility node.

2 Base pan · the guide rails

Four of the pan's own bolts come out; the rails go under them; M8 × 16 go back in.



- 1× Guide rail L, channel open to the left
- 1× Guide rail R, channel open to the right
- 4× M8 × 16 button head, the old washer

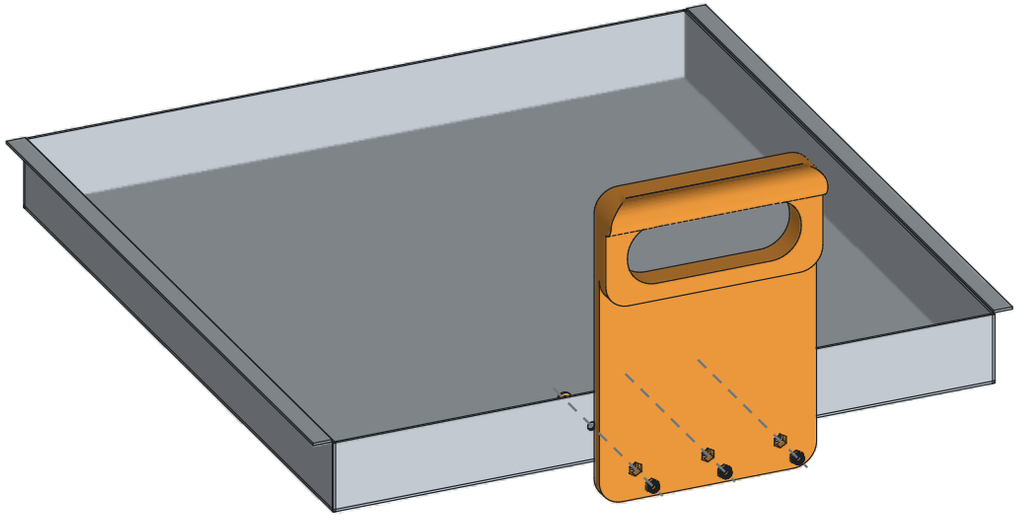
5 mm hex

x 150 and x 650, front line (y 20) and back line (y 539): the pan's own T-nuts stay in the rails underneath. Keep the M8 × 12s.

Take out the four bolts at x 150 and 650 on the pan's front and back lines; keep their washers, the T-nuts stay put in the frame. Lay each rail on the pan with its bottom flange over the two holes, the stop tab at the back, the channel opening away from the middle. Washer, bolt through the rail's flange and the pan into the old nut; snug both, check the rails are parallel (460 between the webs), then tighten.

3 The tray · its handle

On the bench. Three M5 × 10 from inside the tray, three nylocs pressed into the handle's pockets.



- 1× Handle, printed
- 3× M5 × 10 button head
- 3× M5 nyloc

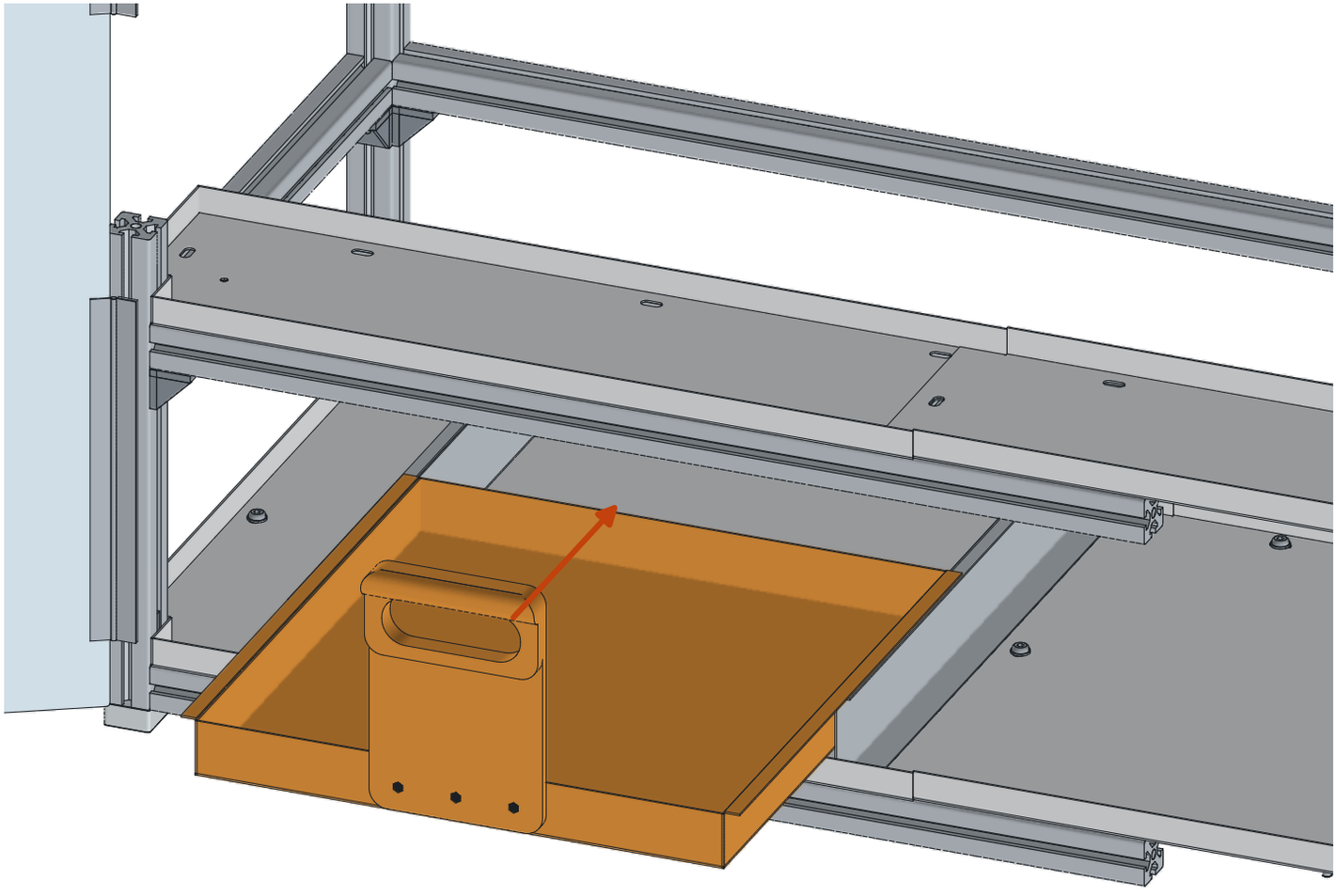
3 mm hex

Press each nyloc into its hex pocket, nylon side out: the pocket holds it while you turn the screw from inside the tray.

The handle stands up off the tray's front wall: its slot is 120 wide and 42 tall, a whole hand through it, the bar rounded over the top. Its flat back goes against the wall's outside, its foot just clear of the bend. Screws from inside, heads on the tray's inside face; tighten until the handle does not rock, not until the plastic whitens.

4 The tray · slide it in

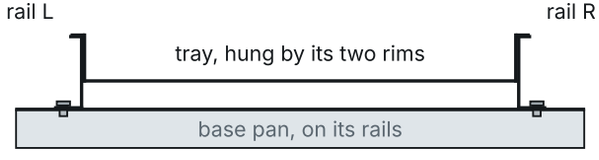
Bottom-left door open. Rims on the rails, push it home to the stops.



1× Dead-leaf tray with its handle

Floor 5.0 over the pan's front lip,
rims on the rails' top flanges,
1.5 of play each side.
Only the left door need be open:
the tray clears the astragal by
37.0.

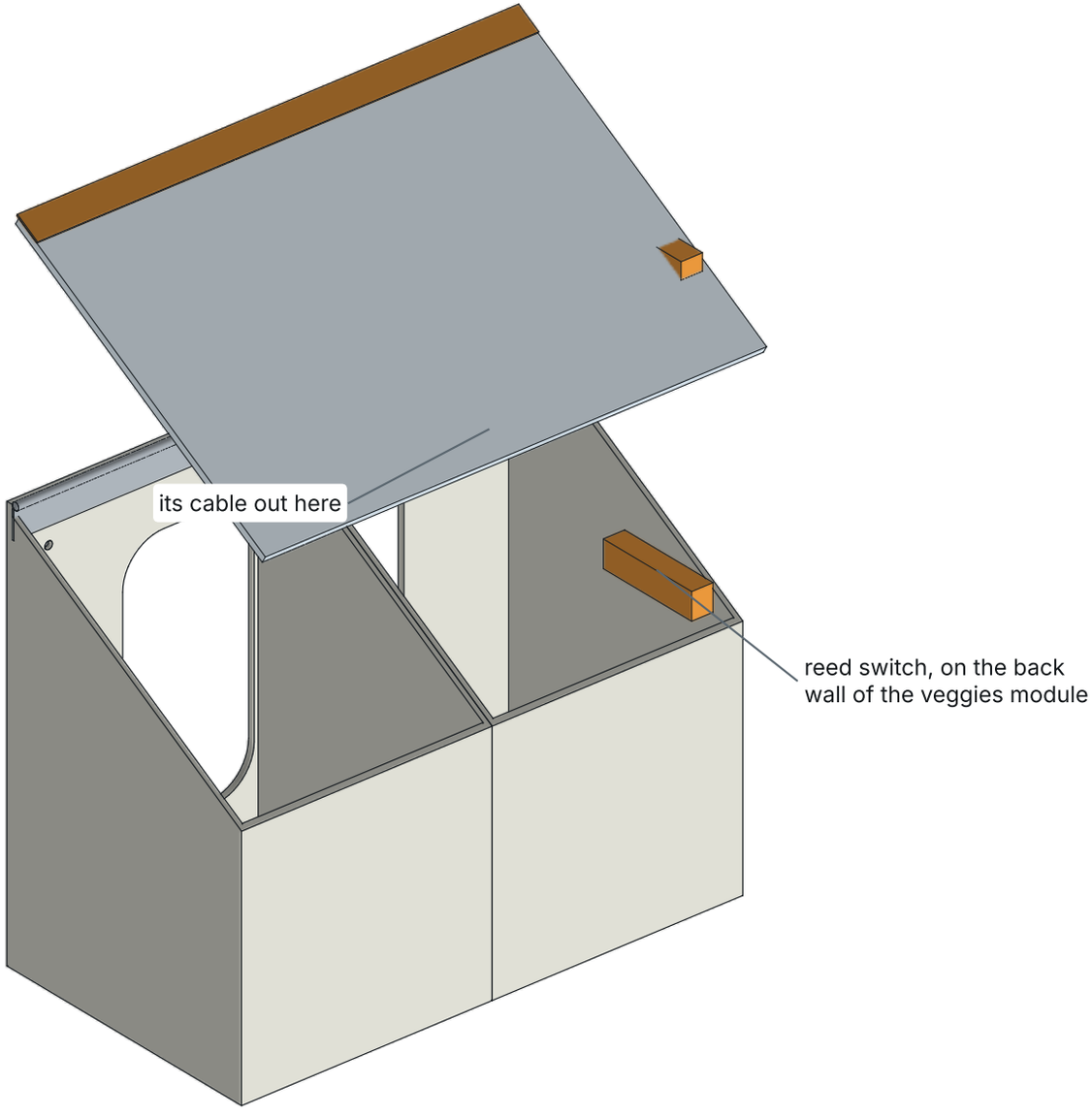
across, from the front



Hold the tray by its handle, tip the back up a little, and set both rims on the rails' top flanges at the front. Push it back until its back wall touches the stop tabs. To empty it, pull it straight out by the handle; past half way the rails no longer balance it, so take its weight and lift it clear. The leak sensor stays where it was, on the pan left of the left rail.

5 The feeder · on the bench

The switch inside the veggies module, its magnet under the lid, the hinge's lid leaf on VHB.



- 1× Reed switch 7939WG, its own adhesive
- 1× Its magnet, under the lid
- 1× Piano hinge 1 ft: lid leaf on VHB

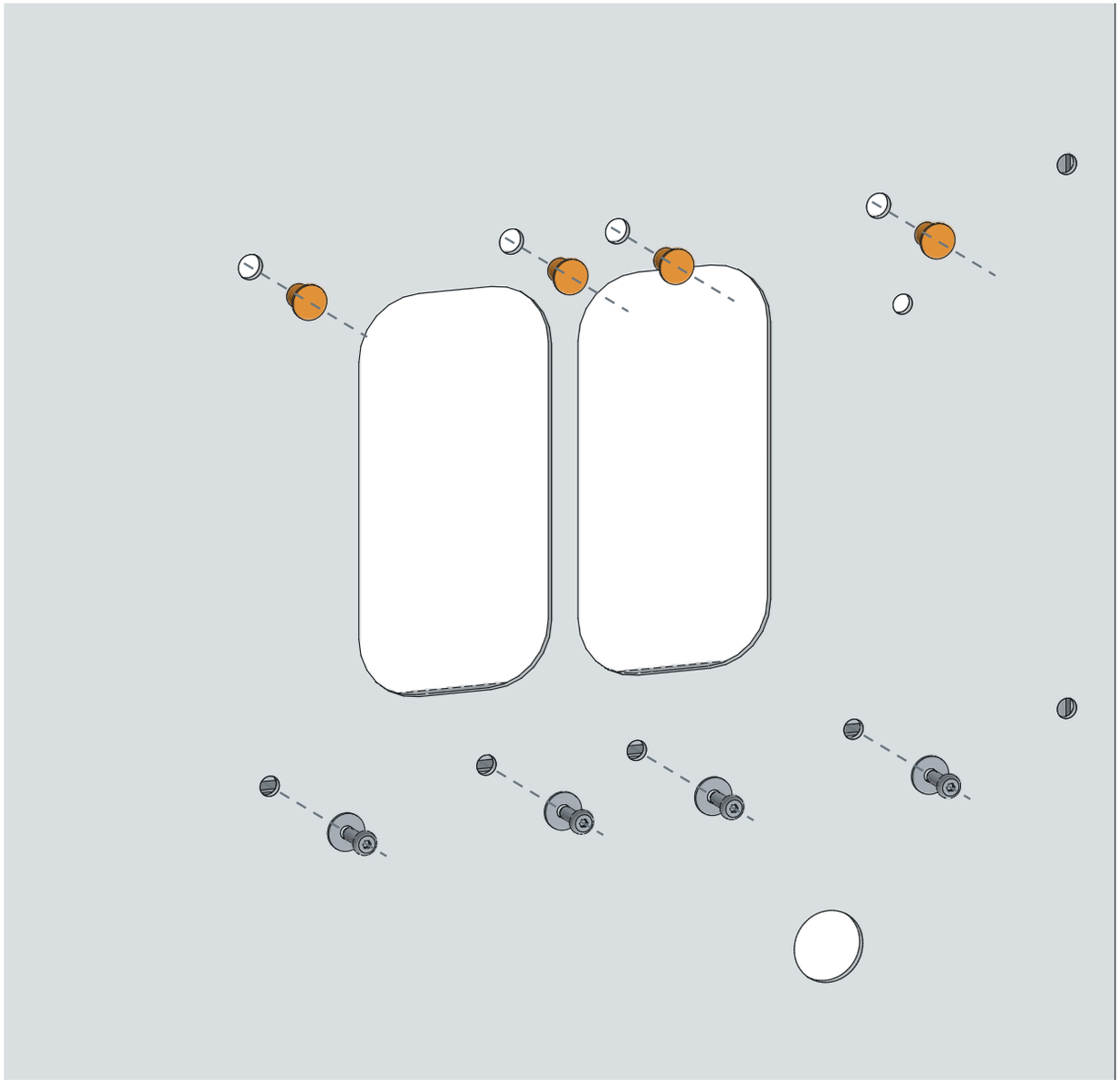
isopropyl · utility knife

Switch and magnet 5.0 apart with the lid down; it works to 32. The magnet's centre: 131 down the lid from its hinge edge, 10 in from its back end.

Stand the modules side by side on their floors, leaves in front (toward the cabinet's front), veggies behind. Clean and stick the switch on the inside of the veggies module's back wall, its top 15 under the lid line, and push its cable out through the 8 hole in the back plate. On the lid: clean it, the magnet where the note says, on the underside; then one VHB strip on the hinge's lid leaf, the knuckle on the lid's top edge. Press each for 30 s.

6 Right panel · well nuts in, temporary screws out

Four well nuts into the 10 holes over the openings; the four green M5 × 16 come out.



4× M4 rubber well nut, into a 10 hole

4× M5 × 16 + washer: out, keep the washers

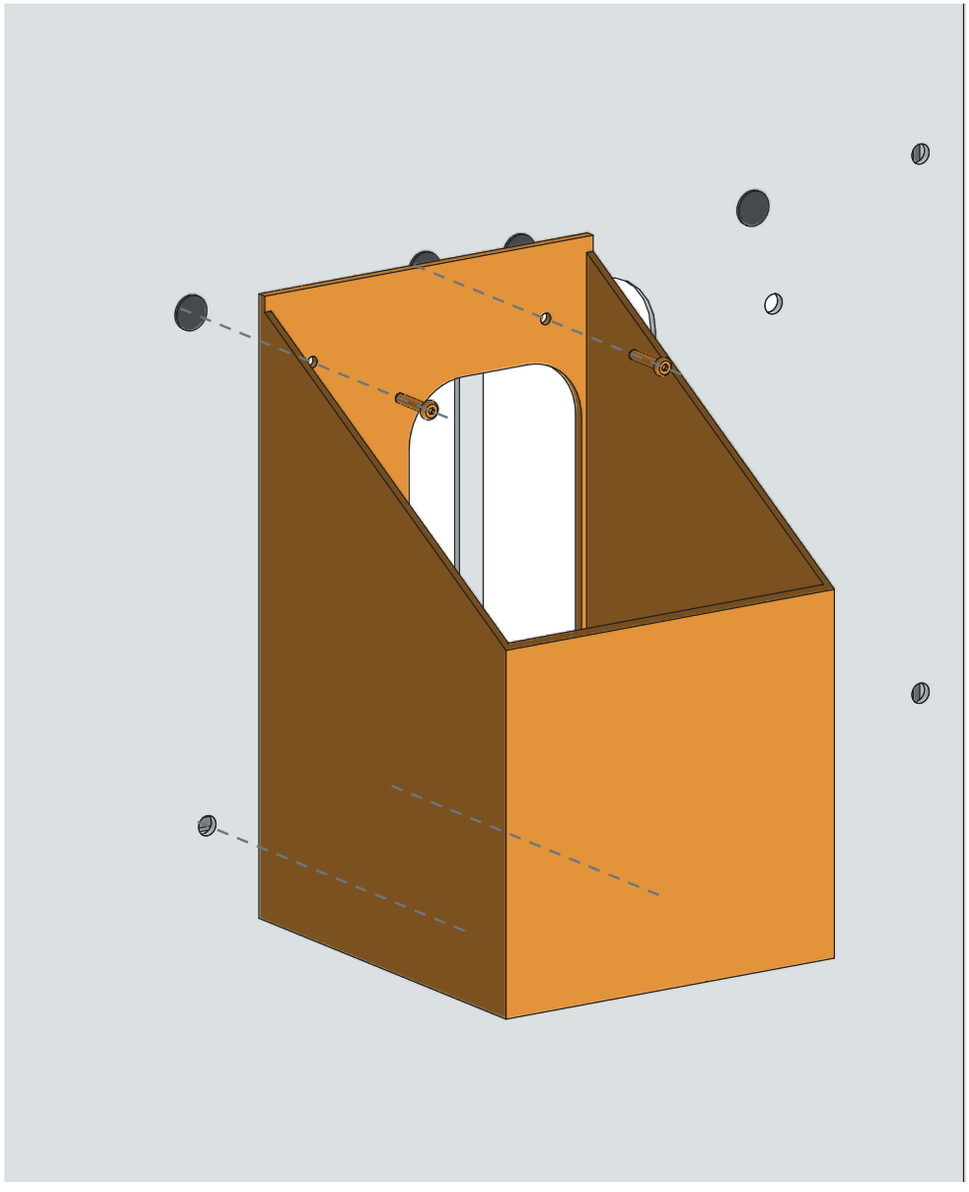
3 mm hex

Take the M5s out one at a time. Their T-nuts stay in the slot behind the panel: the panel covers the slot, so a nut can turn loose but not fall out. The M5 × 20s find them.

Push each well nut into its hole from outside until its flange sits on the panel. Then the four M5 screws on tier B's rail line, under the openings: unscrew and keep their DIN 9021 washers for the feeder. The panel stays put: every other screw still holds it, and its tape is stuck.

7 Feeder · hang the leaves module

Its back plate on the panel, two M5 × 20 into tier B's rail, two M4 × 20 into the well nuts.



- 1× Feeder module, leaves
- 2× M5 × 20 + DIN 9021 washer
- 2× M4 × 20 button head

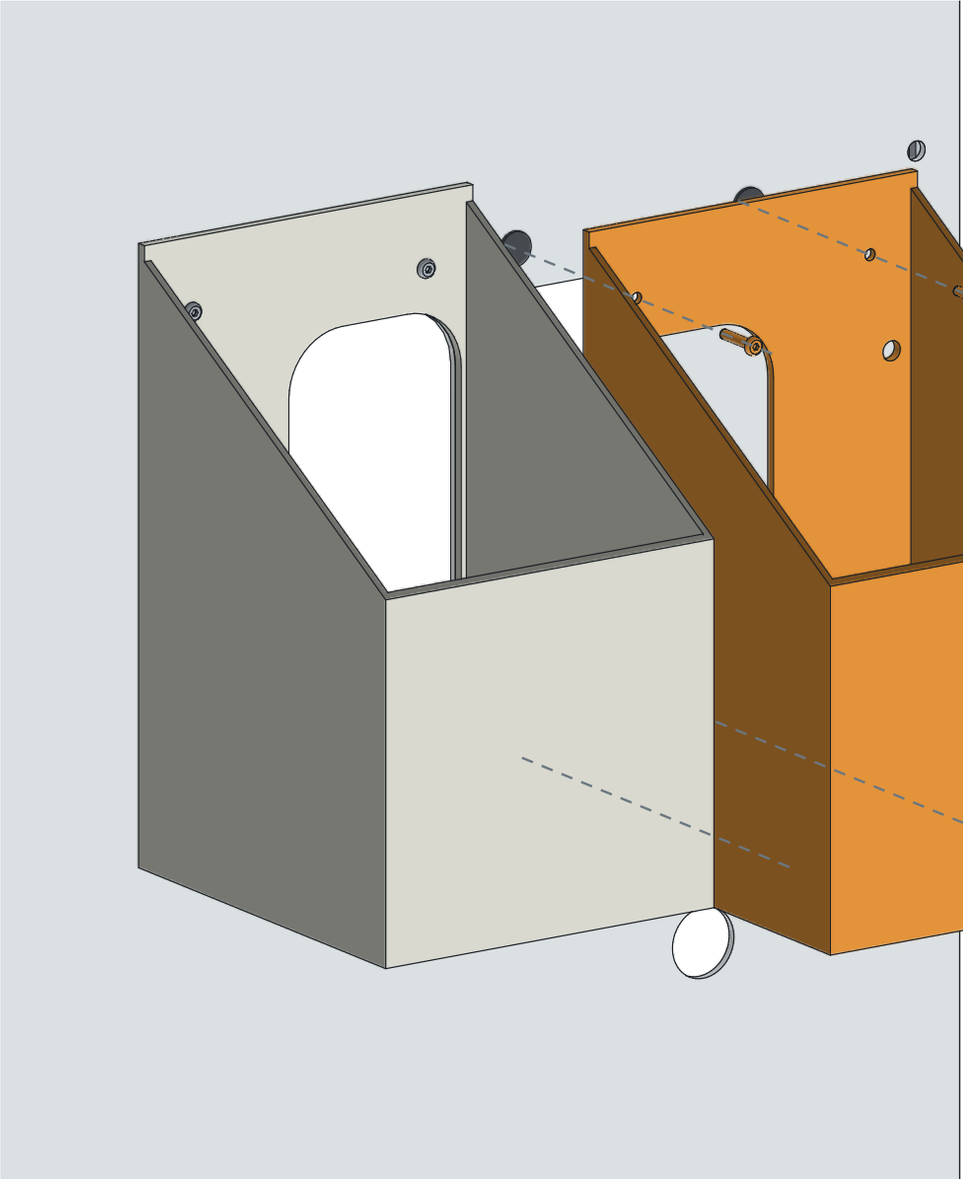
3 mm hex · 2.5 mm hex

Leaves in front. Its front wall lines up with nothing on the panel: the four holes place it.

A helper holds the module against the panel, its back plate's holes over the panel's. Start both M5 × 20s through their washers into the T-nuts behind (turn until each nut swings across and grips), then the two M4 × 20s into the well nuts: they draw the rubber up behind the panel. Snug all four, check the opening in the back plate sits square on the one in the panel, then tighten: the M5s firmly, the M4s until the rubber bulges.

8 Feeder · hang the veggies module

Its back plate on the panel, two M5 × 20 into tier B's rail, two M4 × 20 into the well nuts.



- 1× Feeder module, veggies
- 2× M5 × 20 + DIN 9021 washer
- 2× M4 × 20 button head

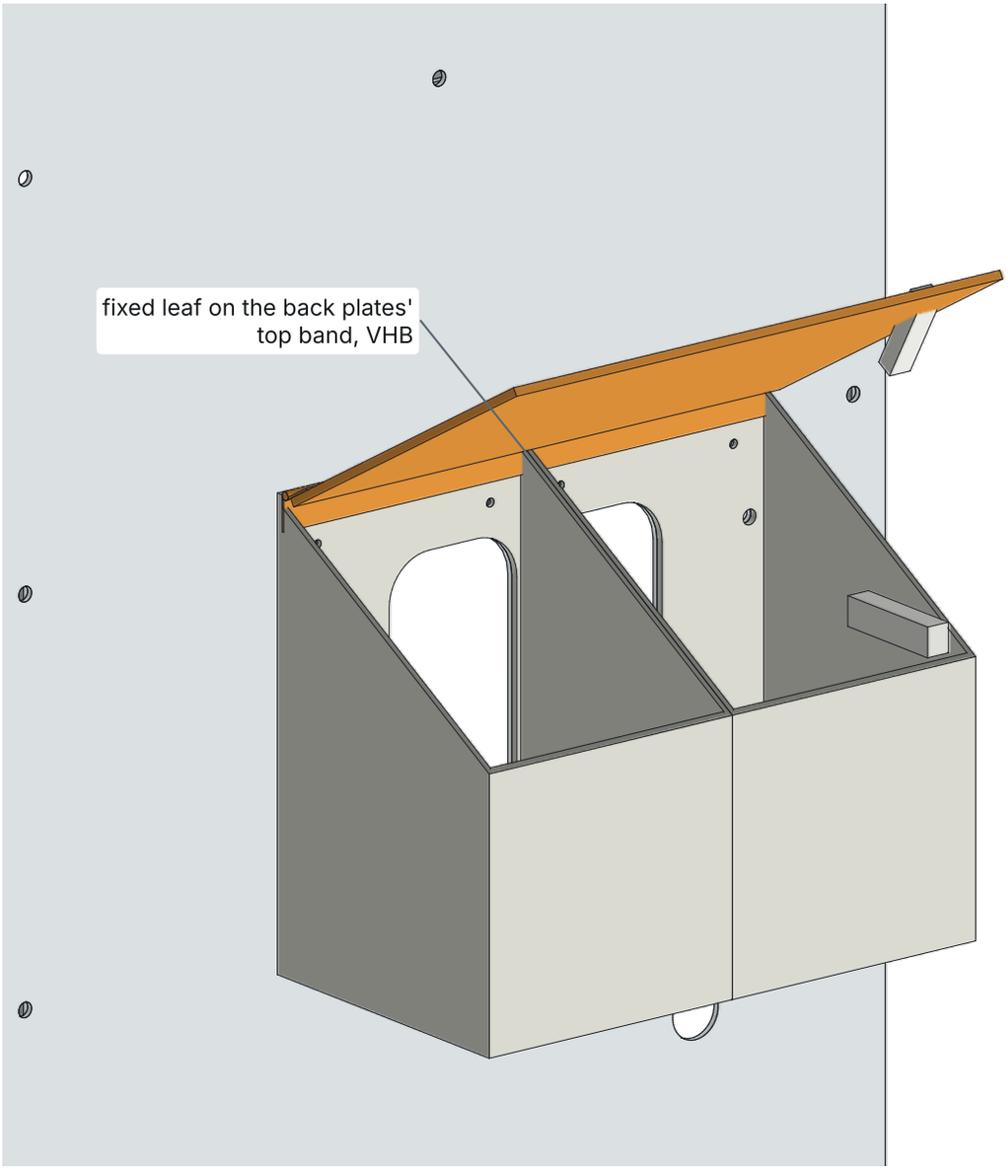
3 mm hex · 2.5 mm hex

Feed the switch's cable through the panel's 8 hole first, then bring the module up to the panel. Its front wall meets the leaves module's back wall: the divider.

Push the reed switch's cable through the 8 hole in the panel behind the module's back plate, and pull the slack into the cabinet as the module goes on. The same four screws as the leaves module. The two modules' walls meet in the middle and make the 6 divider; the lid's hinge needs both back plates' top edges level, so check them with a straightedge before you tighten.

9 Feeder · the lid

The hinge's other leaf on VHB across both back plates. The lid comes down on the side walls.



1× Lid, with its leaf and magnet

1× VHB 4905 strip, the fixed leaf

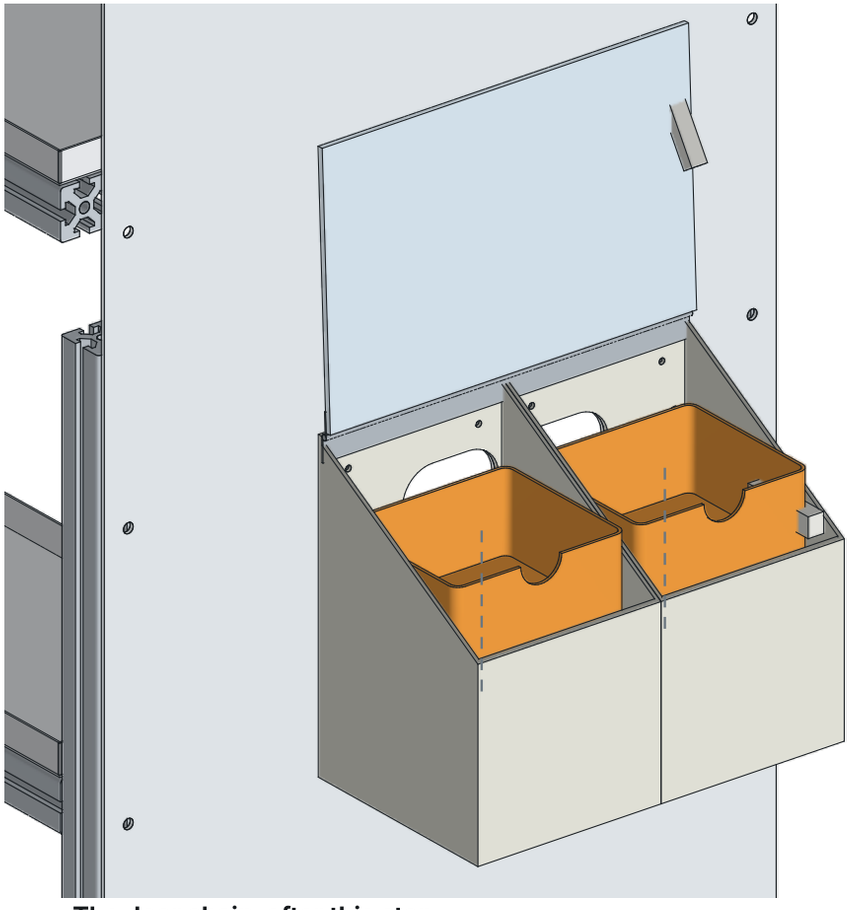
isopropyl · utility knife

Lid 304.8 × 206.3, falling 30° away from the panel. Lifted, it turns past upright and leans on the panel by itself.

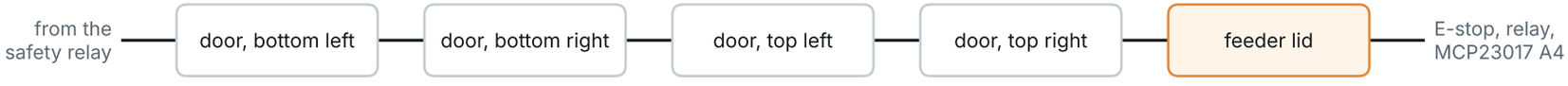
Clean the outer face of both back plates' top band. Hold the lid closed on the side walls, its hinge edge against the panel, and mark where the fixed leaf lands; open it, put VHB on the fixed leaf, close the lid and press the leaf onto both back plates along its length. Leave it closed for an hour before you lift it: VHB takes most of its strength in that time and all of it in a day.

10 Trays in, the switch into the door chain

The lid joins the four door switches in series: lift it and the arm loses power.



The door chain, after this step



- 2× Feeder tray, leaves and veggies
- 2× Lever connector, Wago 221-412

`side cutters · wire stripper`

Each tray's thumb notch faces you. The trays are the same part; mark one L and one V.

Inside the cabinet, run the switch's cable from the panel's 8 hole down the back right post to the utility node, clipped every 300. At the door chain, open the loop at one end and put the lid switch in it with the two lever connectors: five switches in series. The Doors sensor in Home Assistant now reads open when any door or the lid is open, and the safety relay cuts the gantry and the arm.

11 Before the first harvest

The checks that catch the usual faults, with nothing in the trays yet.

- [ ] **Tray**
  Slides out with the left door alone and back to the stops; nothing scrapes; the leak sensor still reads dry.
- [ ] **Lid**
  Lifts, turns past upright, leans on the panel; drops shut on its own and sits on all four walls.
- [ ] **Switch**
  Lid open: Doors reads open, the relay drops, the arm will not move. Lid shut: it resets.
- [ ] **Air**
  Fans on: a tissue at the lid's gap barely moves. The feeder is part of the cabinet's air.
- [ ] **Drops**
  Commissioning: jog to each pose in drops.yaml at 20 % speed, lid shut. The jaws pass the openings' middles.
- [ ] **Software**
  The Herb Snip sends leaves and veggies to the feeder, dead leaves to the tray, five drop points in turn.

Empty the feeder's trays as often as you harvest; wash them like any food container (no dishwasher for PLA). The dead-leaf tray holds about 10 L: empty it into the compost when the leaves reach its rims. Never reach into the feeder from inside the cabinet with the robot powered.